Week 12 16–20 November 2026

Subject: Computer science Unit: Network protocols and Term 1 revision Grade: 11DP SL

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1 Network protocols 60 minutes	A2.1.4 Describe TCP and UDP transport functions, and HTTP, HTTPS and DHCP application functions. Choose suitable protocols for reliable transfer, live communication, secure web access and automatic configuration; explain how they work together.	5 min: retrieve device roles. 15 min: model protocols and TCP/UDP trade-offs. 20 min: sort scenarios; trace DHCP configuration followed by secure web access. 15 min: individual Paper 1-style responses and peer review. 5 min: exit check. Support: protocol-purpose/layer grid and sentence stems. Extend: explain why HTTPS does not guarantee trustworthy content.	Note A2.1.4; IB guide p. 31; protocol cards; request/response and DHCP diagrams. Homework: complete a short five-protocol revision table and justify TCP/UDP choices in two scenarios.
Lesson 2 Exam revision 1 Hardware, data and logic 60 minutes	Revise A1.1.1–.2, .4–.5, .7–.8 and A1.2.1–.5. Explain CPU/GPU, registers, buses, memory and storage choices. Apply conversions, encoding and compression. Complete truth tables, construct circuits and simplify logic using algebra/K-maps.	5 min: retrieval quiz. 30 min: three 10-minute stations—(1) hardware/memory/FDE/storage; (2) conversions, encoding/RLE; (3) truth tables, diagrams and simplification. 15 min: individual timed mixed practice. 10 min: mark, correct and record a revision target. Support: worked models, grids and glossary. Extend: justify choices and verify equivalence.	Concise study guide pp. 1–8; relevant A1.1/A1.2 notes in Toddle Class flow; teacher-prepared parallel questions and answers. Preparation: reread these pages. Homework: correct errors and practise the weakest topic.
Lesson 3 Exam revision 2 OS, scheduling and networks 60 minutes	Revise A1.3.1–.4, A2.1.1–.4 and A1.1.9. Trace FCFS/round robin; calculate mean FCFS waiting time and compare first-response times. Review OS functions, polling/interrupts, cloud services, infrastructure, devices and protocols. Apply ideas to a case study.	5 min: retrieval. 20 min: scheduling trace/calculation clinic plus OS and polling checks. 20 min: small-group case linking networks, cloud/edge and protocols. 10 min: individual timed case responses. 5 min: feedback and final revision target. Support: ready-queue templates, comparison grids and sentence stems. Extend: justify a trade-off and address a counterargument.	Concise guide pp. 4, 9–12; A1.3/A2.1 and cloud-service notes in Toddle Class flow; separate pre-seen case study; parallel practice questions. Homework: review targets, key terms and case-study preparation challenges.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير			

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Technology Integration دمج التكنولوجيا	Use protocol scenario cards, a shared revision quiz and digital/paper worksheets. Project worked scheduling traces and logic examples after students attempt them independently. Provide accessible paper alternatives.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: binary/hexadecimal conversions, mean waiting time and Boolean reasoning. Physics: hardware and signal transmission. Citizenship: privacy and responsible network use.
External Assessment Integration دمج التقييمات الخارجية	Prepare for a 75-minute, 50-mark SL Paper 1-style term exam: Section A 38 marks and Section B 12. Practise command terms, showing working and evidence-based case responses. Use timed parallel tasks and point-by-point feedback, including consistent error-carried-forward where appropriate.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking: diagnose errors and compare alternatives. Communication: explain reasoning to a partner. Self-management: set a revision priority after each session. Use low-stakes retrieval, supportive feedback and realistic exam pacing to build confidence.
Standards المعايير	IB Computer science guide, first assessment 2027, pp. 28–31. New teaching: A2.1.4. Revision: all supplied SL objectives in A1.1, A1.2, A1.3 and A2.1, plus A1.1.9 cloud services. Exam structure: guide pp. 46–49.
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	TOK: How can we tell whether a technically secure source is also a reliable source of knowledge? Revisit HTTPS versus trustworthy content. In revision, ask what evidence supports a technical recommendation.
Success Criteria معايير النجاح (Measurable)	MUST: Identify the purposes of TCP, UDP, HTTP, HTTPS and DHCP; complete core retrieval/conversion tasks and state one revision priority. SHOULD: Explain two protocol choices; complete a combined-circuit table/diagram, a FCFS/RR trace and a mean waiting-time calculation; justify a case response. COULD: Independently simplify and verify logic, distinguish response from waiting time, and evaluate alternatives using scenario evidence.
National Identity Domain مجالات الربط بالهوية الوطنية	Context: use hypothetical UAE school networks and connected-campus services to apply protocol, infrastructure and resource-management ideas.
	Values: reinforce privacy, fair access to shared resources, academic integrity and responsible study practices.
	Application: compare local/edge and remote/cloud processing for a hypothetical school or smart-building system, considering latency, resource limits and connectivity.